Rotaract Club of Xiamen University Malaysia

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	**HTL**
Project / Solution Name	**Quora Chat**
Name (Leader)	**Tng Kar Ming**
Email & Phone (Leader)	**karming0425@gmail.com / +60173553022**
University Affiliation(s)	**Asia Pacific University of Technology & Innovation**
Live Prototype URL	https://quora-chat.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/jasonwong1025/quora-chat

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Quora Chat is a web-based AI-assisted customer service and quotation management system designed for integration with the Orbyte SME ecosystem.

The application provides two interfaces: a customer-facing WhatsApp-style chat and a company-facing Orbyte workspace. Customer enquiries are processed through a generative AI service that identifies requirements, asks clarifying questions, and retrieves relevant public business knowledge.

Structured requirements are stored in PostgreSQL and displayed in the Quotation Inbox for employee review. The quotation module applies predefined pricing rules or staff-provided estimates, allowing authorized employees to approve, reject, or revise quotations before generating PDF documents and customer eSign links.

Following customer acceptance, a background worker creates an operational job and prepares a structured handover containing the approved requirements, scope, and customer commitments. Internal Team Chat supports continued collaboration with an AI assistant that references authorized business knowledge.

The backend uses deterministic rules for pricing, approvals, state transitions, and data validation. Replit hosts the deployed application and provides managed AI integration.

WhatsApp, Orbyte services, email delivery, and payment interactions are simulated for the hackathon demonstration.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Typescript 6.x	Primary application language and business logic	Apache-2.0
React 19.x	Interactive user interface	MIT
TanStack Start 1.x	Full-stack application and routing	MIT
Hono 4.x	Backend HTTP Service	MIT
oRPC 1.x	Type-safe client-server communication	MIT
Drizzle ORM 0.45x	Database schema and queries	Apache-2.0
PostgreSQL 16	Relational data persistence	PostgreSQl License
Tailwind CSS 4.x	Responsive interface styling	MIT
shadcn/ui	Reusable UI components	MIT
Zod 4.x	Structured AI-output and input validation	MIT
React PDF 4.x	Quotation and billing PDF Generation	MIT
OpenAI JavaScript SDK 7.x	Calling supported AI model APIs	Apache-2.0
Node.js 22	Server-side JavaScript runtime	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI API via Replit AI Integrations	Multilingual conversations, requirement extraction, summarization, and AI-generated responses in the deployed application.	Replit-managed; usage-based
OpenRouter API	OpenAI-compatible model access during local development and testing	Usage-based; team account

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Reserved VM Deployment	Hosts the web application and background workflow worker	Reserved VM
Replit PostgreSQL Database	Stores customers, requirements, quotations, approvals, jobs, messages, and audit records	Replit-managed PostgreSQL
Replit AI Integrations	Provides managed API access to the AI model	Usage-based
GitHub	Source code management and version history	
OpenRouter	Alternative AI inference provider for local development	Usage-based

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GPT-5 mini	OpenAI via Replit AI Integrations	Pre-trained model; conversational AI, structured requirement extraction, summarization, drafting	OpenAI API service terms
GPT-5 mini	OpenAI via OpenRouter	Pre-trained model; local development and testing	OpenRouter and model-provider terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic SME Business Dataset	Team-created; project repository	Synthetic	Team-owned	N
Synthetic Customer Conversation Scenarios	Team-created; project repository	Synthetic	Team-owned	N
Product Catalogues and Pricing Rules	Team-created; project repository	Synthetic	Team-owned	N
Business Knowledge and SOP Examples	Team-created; project repository	Synthetic	Team-owned	N
Workflow and Evaluation Test Cases	Team-created; project repository	Synthetic	Team-owned	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	AI-assisted implementation, application configuration, debugging, and deployment support	Major
Claude / Claude Code	Code implementation, refactoring, debugging, and technical improvements	Major
OpenAI Codex	Code implementation, code review, testing, and debugging	Major
Kiro	AI-assisted implementation and development tasks	Moderate
Google Gemini	Brainstorming, research, problem exploration, and solution planning	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team used AI development assistants to support implementation, debugging, research, and documentation while retaining responsibility for the project's requirements, architecture, and technical decisions. Generated code was reviewed against the intended workflow and checked through TypeScript validation, linting, unit tests, integration tests, and manual application testing. We examined important business rules involving requirement consistency, quotation approvals, access restrictions, and workflow reliability. AI-generated suggestions were adapted when necessary, rather than accepted without review. The team remains responsible for the accuracy, security, originality, licensing, and maintainability of the submitted work.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core	Application development, project configuration, and AI-assisted development
Replit	Reserved VM Hosting	Hosting the deployed prototype and persistent background worker
Replit	Replit AI Integration	Accessing the OpenAI model for application inference

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the deployment environment and managed AI services used for the hackathon prototype. It simplifies application hosting, database connectivity, and access to generative AI capabilities. These resources are important for the current implementation and satisfy the event's sponsor usage requirement. However, Quora Chat's application architecture is not permanently dependent on Replit. Its web services, PostgreSQL database, and AI API connections could be migrated to compatible infrastructure. RunPod GPU resources were not required because the current solution uses pre-trained AI models through hosted APIs rather than performing local model training or GPU inference.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Quora Chat currently operates as a hackathon prototype using synthetic customer profiles, conversations, employee records, product information, quotations, and business documents. No real personal or customer records are intentionally used.

The application stores conversation histories, extracted requirements, quotation records, approvals, job details, and activity logs in PostgreSQL. Relevant conversation context and authorized knowledge may be transmitted to the configured third-party AI provider for processing.

Application credentials are supplied through environment variables rather than intentionally embedded in source code. Public and internal knowledge are separated through application-level access controls.

The prototype uses simulated employee identities and does not provide production-grade authentication. Consequently, it should not be used for real customer data or sensitive operational records.

Data is retained in the demonstration database until manually reset or deleted. Production deployment would require defined retention periods, authenticated user access, stronger authorization, and appropriate compliance with Malaysia's Personal Data Protection Act (PDPA).

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The generative AI model may misunderstand multilingual messages, omit requirements, produce incorrect summaries, or generate unsupported information. Quora Chat mitigates these risks through structured output validation, predefined requirement questions, evidence references, clarification prompts, and human review.

AI does not independently determine commercial pricing or authorize consequential business actions. Deterministic application logic governs pricing, quotation approvals, requirement versioning, and workflow transitions.

The solution depends on external AI APIs, which may experience latency, outages, rate limits, or usage costs. Background processing incorporates retry mechanisms, while failed tasks may require manual intervention.

Other limitations include simulated WhatsApp and Orbyte integrations, demonstration-only user identity selection, and knowledge retrieval that does not yet extract full document attachment content.

Further evaluation is required to establish multilingual accuracy, usability, and reliability under real SME workloads.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Tng Kar Ming

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24 October 2026